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# Introduction to Embedded Systems & IoT Device Design

🤖 **Embedded Systems Task-1**

## Objective

The goal of this task is to help you understand the **fundamentals of embedded systems** and apply them by creating a **small IoT-based project simulation**.

By completing this task, you will:

- Learn what embedded systems are and where they are used.
- Gain **hands-on exposure** to microcontrollers and sensors (real or simulated).
- Create your **first portfolio-ready embedded systems mini project**.

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## 📝 Task Description

You will act as a **Junior Embedded Engineer** in a smart devices company. Your manager has asked you to research **embedded systems concepts** and design a **basic IoT prototype (simulation).**

Your task includes:

1. **Introduction Section (Research)**
    - What is an embedded system?
    - Real-world applications (smart home, automotive, healthcare devices, robotics, etc.).
    - Difference between Microcontroller vs Microprocessor.

2. **Component Study**
    - Research and explain the role of **Microcontrollers (Arduino, ESP32, Raspberry Pi Pico).**
    - Basic sensors (Temperature sensor, Motion sensor, Light sensor).

3. **Mini Project – Smart Sensor System (Simulation Allowed)**
    Choose one of the following:
    - **Smart Temperature Monitor** → Reads temperature values and shows alerts when crossing threshold.
    - **Smart Lighting System** → LED turns ON/OFF based on sensor input.
    - **Motion Detection Alarm** → Motion sensor triggers a buzzer/LED when movement is detected.

👉 You can implement this either:

- On **real hardware** (if available: Arduino UNO, ESP32, sensors, LEDs).
- Or on a **simulator** like Tinkercad Circuits (free, browser-based).

4. **Documentation & Report**
    - Explain the **working principle** of your chosen system.
    - Add **circuit diagram** (Tinkercad auto-generates, or use Fritzing).
    - Provide **code (Arduino/C++)** with explanation.
    - Describe potential **real-world applications** of your mini project.

## 📚 Guidelines & Free Resources

**Learning Basics**

- Introduction to Embedded Systems – NPTEL (Free)
- Arduino Official Getting Started Guide
- Wokwi Simulator (Free Arduino/ESP32 Simulator)
- Tinkercad Circuits – Beginner-friendly circuit builder

### Sample Codes

- Arduino LED Blink (Hello World of Embedded):

```cpp
void setup() {
  pinMode(13, OUTPUT);
}
void loop() {
  digitalWrite(13, HIGH);
  delay(1000);
  digitalWrite(13, LOW);
  delay(1000);
}
```

- Modify this to control **LED with a sensor** (e.g., motion/light sensor).

### Tools Needed

- Hardware: Arduino UNO (optional, if student has access).
- Software: Arduino IDE (Free Download).
- Alternative: Online Simulators (Tinkercad, Wokwi).

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## 📤 Deliverables

- A **project report (PDF/DOCX)** containing:
    - Introduction to Embedded Systems.
    - Your selected project details (circuit diagram, explanation, applications).
    - Source code with comments.
- (Optional) A **short video demo** if you test on hardware/simulator.